



Did You Know?

SQL stands for Structured Query Language. It is the standard language used to communicate with relational databases.



What is SQL?

SQL (Structured Query Language) is a programming language used to create, retrieve, update and delete data stored in a database.



Why do we use SQL?

- Store data

- Retrieve records
- Update information
- Delete unwanted data
- Manage databases



Common SQL Commands

Command	Purpose
SELECT	Retrieve data
INSERT	Add new records
UPDATE	Modify existing records
DELETE	Remove records
WHERE	Filter records
JOIN	Combine multiple tables



SELECT Example

```
SELECT * FROM Students;
```



INSERT Example

```
INSERT INTO Students  
VALUES(1, 'Noor', 'BSCS');
```



UPDATE Example

```
UPDATE Students  
SET Department='CS'  
WHERE ID=1;
```



DELETE Example

```
DELETE FROM Students  
WHERE ID=1;
```



WHERE Example

```
SELECT *  
FROM Students  
WHERE Department='BSCS';
```



JOIN Example

```
SELECT Students.Name,  
Courses.CourseName
```

```
FROM Students
```

```
JOIN Courses
```

```
ON Students.ID=Courses.StudentID;
```

Advantages of SQL

- Easy to learn
- Fast data retrieval
- Standard database language
- Supports large databases
- Secure data management

Common Mistakes

- Missing WHERE clause
- Wrong table name

- Typing incorrect SQL keywords
- Using DELETE without WHERE



Interview Questions

Q1: What is SQL?

SQL is the language used to manage relational databases.

Q2: Which command retrieves data?

SELECT

Q3: Which command adds new records?

INSERT

Q4: Which command combines multiple tables?

JOIN

Quick Revision

SELECT Read Data

INSERT Add Data

UPDATE Modify Data

DELETE Remove Data

JOIN Combine Tables



Download Database Notes

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